



MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES
BSE4104: EMERGING TRENDS IN SOFTWARE ENGINEERING

AI-NATIVE & AGENTIC ENGINEERING CAPSTONE: WEEK 3 SUBMISSION

WEEK 3 RAG EVALUATION

PROJECT: UNIVERSITY STUDENT-SUPPORT CASE AGENT

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14-18 SEPTEMBER 2026

Week 3 RAG Evaluation

Run date: 19 September 2026

Harness: tests/test_rag_evaluation.py (R1-R15)

Index: existing local ChromaDB index, persisted with ONNX embeddings

Model result: local deterministic evaluation passes; live Gemini evaluation requires RAG_EVAL_LIVE=1 and available API quota

Status table

The table records the current evidence; it is not presented as a passing final evaluation. Source rankings are retrieval-only observations from the existing index.

Commands and results

- `python -m pytest -q tests/test_retrieval.py`: **10 passed**.
- `python -m pytest -q tests/test_evaluation.py`: **10 passed** using the documented mock harness.
- `python -m pytest -q tests/test_rag_evaluation.py`: **15 passed** with the default local deterministic model stub. Set `RAG_EVAL_LIVE=1` to run against Gemini; earlier live runs were affected by HTTP 429 quota limits.

The offline evaluation is complete and reproducible. A separate live-model run remains useful for recording natural-language answer quality when Gemini quota is available.